

Student 20 **Krithick Rosan M** 192524206RD

5 / 5 submitted

Q1. Program based on Classes and Objects

Score: 3.33 TC: 2/3 passed

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    void displayDetails(){
        System.out.println("Roll No: "+ rollNo);
        System.out.println("Name: "+ name);
        System.out.println("Marks: "+ marks);
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Student s = new Student();
        s.rollNo=sc.nextInt();
        sc.nextLine();
        s.name=sc.nextLine();
        s.marks=sc.nextDouble();
        s.displayDetails();
        sc.close();
    }
}
```

INPUT

```
2
Krithick
98.5
```

OUTPUT

```
Roll No: 2
Name: Krithick
Marks: 98.5
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s=new Student();
        s.setRollNo(sc.nextInt());
        sc.nextLine();
        s.setName(sc.nextLine());
        s.setMarks(sc.nextDouble());
        s.displayDetails();
    }
}
class Student{
    private int rollNo;
    private String name;
    private double marks;
    public void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    public void setName(String name){
        this.name=name;
    }
    public void setMarks(double marks){
        this.marks=marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
}
```

INPUT

193
Eshwar
77.2

OUTPUT

Roll No: 193
Name: Eshwar
Marks: 77.2

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s1=new Student();
        s1.displayDetails();
        int roll=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        Student s2=new Student(roll, name, marks);
        s2.displayDetails();
    }
}
class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo=101;
        name="Rahul";
        marks=85.5;
    }
    Student(int rollNo,String name,double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;
    }
    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
}
```

INPUT

```
102
Nidhi
88.9
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 102
Name: Nidhi
Marks: 88.9
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java .util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int id=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double salary=sc.nextDouble();
        Salary s=new Salary();
        s.getEmployeeDetails(id,name,salary);
        s.calculateSalary();
        s.displayDetails();
    }
}
class Employee{
    int empId;
    String empName;
    double basicSalary;
    void getEmployeeDetails(int id,String name,double salary){
        empId=id;
        empName=name;
        basicSalary=salary;
    }
}
class Salary extends Employee{
    double hra,da,netSalary;
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        netSalary=basicSalary+hra+da;
    }
    void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Net Salary: "+netSalary);
    }
}
```

INPUT

101
Rahul
50000

OUTPUT

Employee ID: 101
Employee Name: Rahul
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
Net salary: 65000.0

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int id=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double salary=sc.nextDouble();
        GrossSalary g=new GrossSalary();
        g.getEmployeeDetails(id,name);
        g.getBasicSalary(salary);
        g.calculateSalary();
        g.displayDetails();
    }
}
class Employee{
    int empId;
    String empName;
    void getEmployeeDetails(int id,String name){
        empId=id;
        empName=name;
    }
}
class Salary extends Employee{
    double basicSalary;
    void getBasicSalary(double salary){
        basicSalary=salary;
    }
}
class GrossSalary extends Salary{
    double hra,da,grossSalary;
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        grossSalary=basicSalary+hra+da;
    }
    void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Gross Salary: "+grossSalary);
    }
}
```

INPUT

```
101
Rahul
50000
```

OUTPUT

```
Employee ID: 101
Employee Nmae: Rahul
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
Gross Salary: 65000.0
```